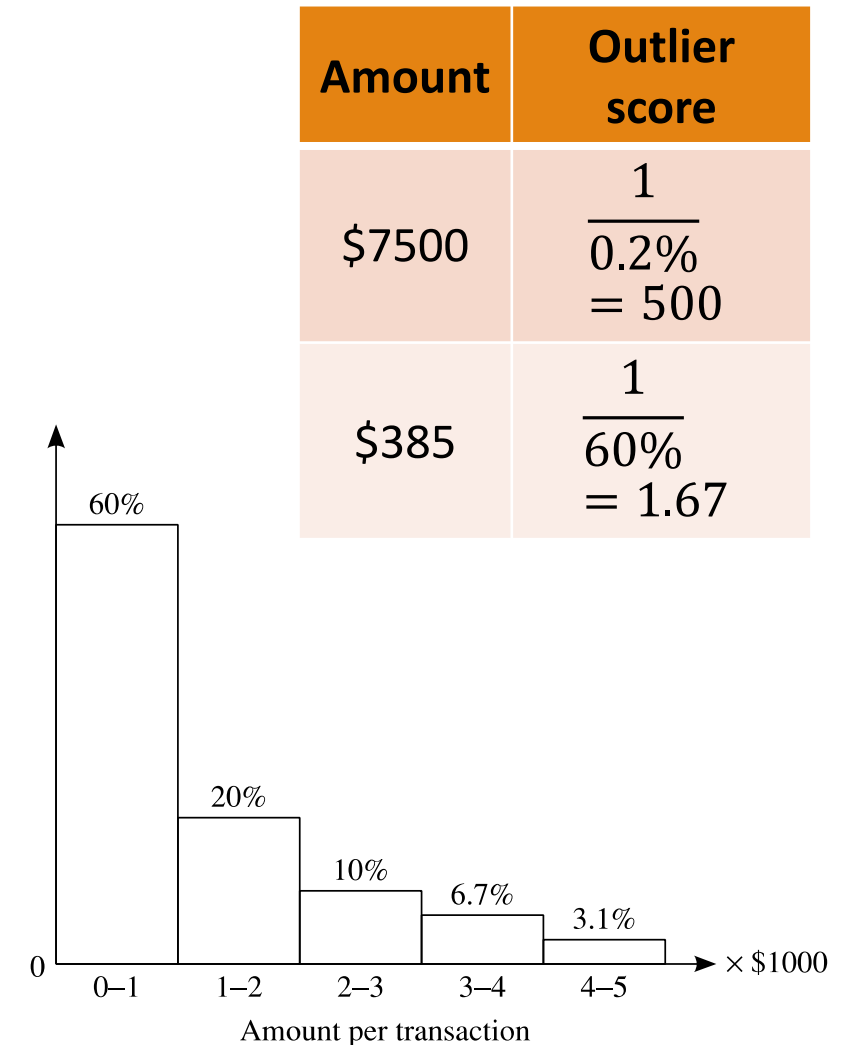


A Nonparametric Method: Using Histogram

- ❑ Construct a histogram using the input data (training data)
- ❑ If the object falls in one of the histogram's bins, the object is regarded as normal
 - ❑ Otherwise, it is considered an outlier
- ❑ Use the histogram to assign an outlier score to an object, such as the reciprocal of the volume of the bin in which the object falls
- ❑ Drawbacks: hard to choose an appropriate bin size



Kernel Density Estimation

- A kernel $K()$ is a non-negative real-valued integrable function that satisfies the two conditions
 - $\int_{-\infty}^{\infty} K(u) du = 1$
 - $K(-u) = K(u)$ for any u
 - A frequently used kernel is the standard Gaussian function with mean 0 and variance 1, that is, $K(x) = \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}$
- Let $x_1, \dots, x_n$ be independent and identically distributed samples of a random variable f
- The kernel density approximation of the probability density $\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x-x_i}{h}\right)$, where $K()$ is a kernel and h is the bandwidth serving as a smoothing parameter